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— MINI PROJECT REVIEW – 1 —

Plant Doctor Chatbot

A Voice-Enabled Retrieval-Augmented Generation (RAG) Chatbot for Agricultural Assistance

Department of Artificial Intelligence

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OUTLINE

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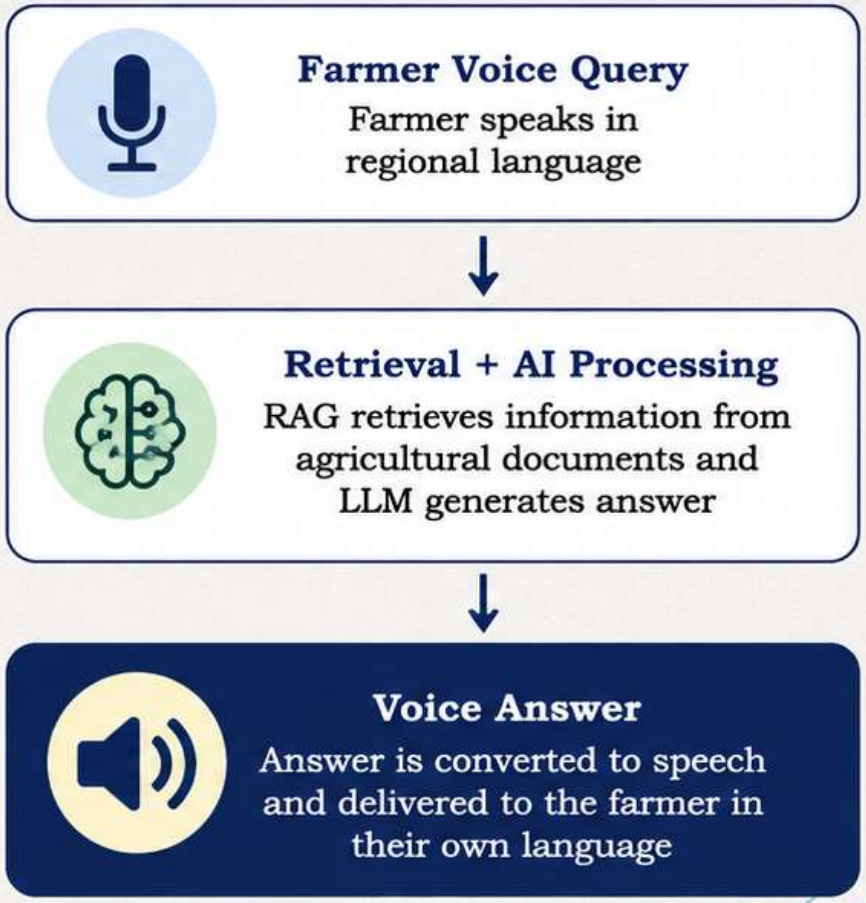
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— OVERVIEW —

Introduction

- Agriculture is the backbone of India, yet farmers face significant challenges in getting **timely** and **accurate** advice for crop problems.
- Lack of **agricultural experts**, **language barriers**, and **scattered information** in documents make it difficult for farmers to find the right solution.
- Existing solutions are either **not in regional languages**, **require typing**, or provide **unreliable** answers.
- Our chatbot uses **voice interaction** and **RAG** (Retrieval-Augmented Generation) to deliver **accurate**, **trustworthy**, and **farmer-friendly** solutions in their own language.

> **Our project** builds a **voice-enabled, multilingual** chatbot that retrieves answers from trusted agricultural documents and provides **accurate guidance** to farmers.



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— MOTIVATION —

The Problem

Lack of Timely Expert Access

Farmers often cannot consult agricultural experts in time, especially in remote areas.

Language Barrier

Most farmers speak regional languages, while agricultural information is mostly available in English.

Scattered & Unreliable Information

Agricultural knowledge is scattered across PDFs, websites, and forums. Farmers find it hard to locate reliable and trustworthy advice.

Digital & Literacy Challenges

Many farmers have low digital literacy and prefer voice interaction over typing. Existing chatbots are text-based and not voice-friendly.

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— GOAL —

Problem Statement

Design and develop a **multilingual, voice-enabled** AI chatbot using **Retrieval-Augmented Generation (RAG)** that retrieves information from trusted agricultural documents and provides **accurate**, **reliable**, and **farmer-friendly** answers in the farmer's own language.

Farmer Voice Query → **RAG + LLM Processing** → **Voice Output**

TARGET USERS

Farmers, Agri-Extension workers, KVK centers, Rural communities

INPUT

Farmer's voice question in regional language

OUTPUT

Accurate answer in text and spoken in local language

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— RELATED WORK —

Literature Survey

Paper / Author (Year)	Methodology Used	Limitation	Our Improvement
AI Research Assistant in Dryland Agriculture (2026)	RAG chatbot over 12,000 validated agriculture publications, semantic embeddings, vector retrieval, open-source LLMs, voice-enabled interface.	Focused more on researchers and extension workers than ordinary farmers; not centered on conversational crop diagnosis.	**Farmer-first design with simpler voice-based Q&A, local-language support, and crop-specific advice.**
Empowering Indian Farmers with Multilingual AI: A Voice-Enabled ChatBot Using Dhenu2 (2025)	Multilingual voice/text pipeline using STT, Dhenu2 for reasoning, and TTS in Indian languages.	More general farming advice; does not emphasize document-grounded retrieval from manuals and guidelines.	**Add RAG with agricultural documents so answers are grounded in verified sources.**
CropGuard: Empowering Agriculture with AI driven Plant Disease Detection Chatbot (2024)	Chatbot framework with AI and deep learning for plant disease detection, frontend in Streamlit, GPT-3.5 Turbo for conversational layer.	More disease-detection oriented; not a full voice-enabled farmer assistant.	**Combine disease guidance with voice input/output and knowledge-base retrieval.**
Smart Plant Monitoring System and Chatbot (2025)	IoT + AI system for plant health monitoring, moisture sensing, pest detection, and chatbot support.	Depends on monitoring hardware and is less focused on natural-language agricultural document Q&A.	**Build a document-driven chatbot that works even without sensors or IoT devices.**
RAG-Powered Chatbot for Agricultural Advice (2026)	Retrieval-Augmented Generation for agricultural advice.	Likely text-first and may not include multilingual voice interaction.	**Add speech-to-text, text-to-speech, and regional language translation for farmers.**

Key takeaway: Existing systems are either not voice-enabled, not domain-specific, or not multilingual.

Our project combines **voice interface, RAG,** and **multilingual support** for accurate agricultural assistance.

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— WHAT IS MISSING —

Research Gaps

G1 **Lack of Voice-First Agricultural Chatbots**

Most existing systems are text-based and do not support real-time voice interaction in local languages, which limits usability among farmers.

G2 **Limited Use of Trusted Agricultural Knowledge**

Many chatbots rely on general information or small datasets, lacking retrieval from verified agricultural documents and publications.

G3 **Insufficient Multilingual Support**

Existing systems support only a few languages or English, leaving a large portion of farmers who speak regional languages underserved.

G4 **No Document-Grounded AI Responses**

Many AI chatbtots generate answers without referring to trusted agricultural documents, so the advice may be incorrect or unsafe for farmers.

HOW WE SOLVE THEM

We build a **voice-enabled RAG** chatbot that retrieves from **verified agricultural documents**, understands **regional languages**, and provides **farmer-friendly**, **crop-specific advice** through conversational AI.

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— WHAT WE WILL DO —

Objectives

1 Collect Agricultural Knowledge Base

Collect trusted agricultural documents from ICAR, KVK, Government portals, agricultural manuals, crop calendars, and pest management guides.

2 Process & Index Documents

Extract text from PDFs, perform OCR (if required), clean the data, split into chunks, generate embeddings, and store them in a Vector Database (FAISS/ChromaDB).

3 Develop RAG Pipeline

Implement a Retrieval-Augmented Generation (RAG) pipeline that retrieves the most relevant document chunks before generating responses using an LLM.

4 Build Voice Interface

Develop a multilingual Speech-to-Text (ASR) and Text-to-Speech (TTS) interface so farmers can ask questions and receive spoken answers.

5 Support Regional Languages

Enable Telugu, Hindi, Tamil, and other regional languages using translation models so farmers can interact naturally in their preferred language.

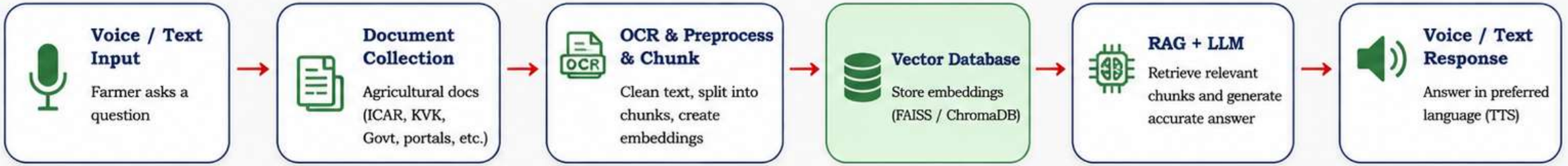
6 Evaluate & Deploy

Evaluate the chatbot for retrieval accuracy, response quality, latency, and multilingual performance, and deploy it as a web/mobile application suitable for rural users.

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— PIPELINE —

Proposed Methodology



PROCESS STEPS

1. Farmer asks a question using voice or types in regional language.
2. The question is converted to text (ASR) and translated if needed.
3. System retrieves relevant information from the Vector Database using embeddings.
4. Retrieved document chunks are passed to LLM with the question using RAG.
5. LLM generates a context-aware answer.
6. Answer is translated to the user's language and delivered via text and voice (TTS).

DATASETS & TOOLS

Documents	ICAR publications, KVK advisories, Government schemes, Crop manuals, Pest management guides, etc.
Preprocessing	PyMuPDF, Tesseract OCR, LangChain Text Splitter
Embeddings	Sentence Transformers (all-MiniLM-L6-v2)
Vector Database	FAISS / ChromaDB
LLM	Llama 3.1 / Mistral 7B (via Ollama)
RAG Framework	LangChain
ASR (Speech-to-Text)	Whisper (OpenAI)
TTS (Text-to-Speech)	Coqui TTS / gTTS
Languages Supported	Telugu, Hindi, Tamil, English + more
Platform	Python, Streamlit (Web)

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— NOVELTY —

Our Contribution

Voice-Based Farmer Assistance

Farmers can ask questions using speech in their regional language.

Speech-to-Text (ASR) and Text-to-Speech (TTS) provide a natural hands-free experience.

Grounded Agricultural Intelligence

Answers are generated using Retrieval-Augmented Generation (RAG).

Information is retrieved from ICAR, KVK, and Government agricultural documents, reducing hallucinations.

Accessible for Every Farmer

Supports Telugu, Hindi, Tamil, English, and other regional languages.

Designed for low-literacy and rural users with a simple voice-first interface.



NOVELTY STATEMENT

Unlike traditional agricultural chatbots, our system combines **multilingual voice interaction, Retrieval-Augmented Generation (RAG)**, and **trusted agricultural knowledge bases** to provide accurate, document-grounded, and farmer-friendly recommendations in regional languages.

Planned Comparison

D1 ICAR + KVK Agricultural Documents (~5000+ documents)

Method	Retrieval Accuracy (Hit@5) ↑	Answer Relevance (F1) ↑	Response Time (s) ↓	Hallucination Rate (%) ↓
Keyword Search	TBD	TBD	TBD	TBD
BM25	TBD	TBD	TBD	TBD
Dense Retrieval (Sentence Transformers)	TBD	TBD	TBD	TBD
Generic RAG (Llama 3.1 8B)	TBD	TBD	TBD	TBD
Proposed RAG System (Ours)	**TBD**	**TBD**	**TBD**	**TBD**

D2 Government Agricultural Schemes / Crop Advisory Documents (~2000+ documents)

Method	Retrieval Accuracy (Hit@5) ↑	Answer Relevance (F1) ↑	Response Time (s) ↓	Hallucination Rate (%) ↓
Keyword Search	TBD	TBD	TBD	TBD
BM25	TBD	TBD	TBD	TBD
Dense Retrieval (Sentence Transformers)	TBD	TBD	TBD	TBD
Generic RAG (Llama 3.1 8B)	TBD	TBD	TBD	TBD
Proposed RAG System (Ours)	**TBD**	**TBD**	**TBD**	**TBD**

- **Higher** Retrieval Accuracy and Answer Relevance are better. **Lower** Response Time and Hallucination Rate are better.
- Results will be reported in **Review 2** after implementation and evaluation.

— IEEE FORMAT —

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Thank You

Questions & Discussion

Department of Artificial Intelligence · Anurag University, Hyderabad

Plant Doctor Chatbot Using RAG and Large Language Models